ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** asexual reproduction	No gametes involved.
2	**(d)** stamen	Anther and filament.
3	**(a)** fruit	Fruit.
4	**(b)** testes	Male organ.
5	**(b)** binary fission	Splits into two.
6	**(b)** carpel (pistil)	Stigma, style, ovary.
7	**(c)** fertilisation	Zygote formation.
8	**(c)** ovaries	Female organ.
9	**(c)** budding	Buds from parent.
10	**(c)** pollen grains	Male gametes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Production of new individuals of the same species (1); Ensures the continuity of species (1).	1+1
12	By spores from sporangia (1); Spores germinate on moist bread (1).	1+1
13	Many eggs and sperms are lost in the water (1); To increase the chance of fertilisation (1).	1+1
14	Tiny reproductive units in a protective coat (1); To increase chances of landing in suitable places (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Nucleus divides into two (1); Cytoplasm divides (1); Two daughter Amoebae form (1).	1+1+1
16	Insect: coloured, fragrant, nectar (1); Wind: small, light pollen, feathery stigma (1); Adaptations (1).	1+1+1
17	Barrier: condom (1); Chemical: pills (1); Surgical: vasectomy/tubectomy (1).	1+1+1
	OR: A bud grows on the parent (1); The bud develops and detaches (1); Forms a new individual (1).	1+1+1
18	External: outside the body, in water, frog (1); Internal: inside the female body, hen, humans (1); Number of eggs differs (1).	1+1+1
	OR: Limited resources (1); Better health of mother and child (1); Better quality of life (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fission (1); Budding (1); Fragmentation (1); Spore formation (1); Vegetative propagation (1).	1+1+1+1+1
	OR: Parents (1); Gametes (1); Variation (1); Speed (1); Examples (1).	1+1+1+1+1
20	Calyx and corolla (1); Stamen and anther (1); Carpel (1); Pollination (1); Fertilisation (1).	1+1+1+1+1
	OR: Condom (1); Pills (1); IUD (copper-T) (1); Surgical methods (1); Reproductive health and hygiene (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that budding is the method of *Amoeba*; *Amoeba* divides by binary fission, whereas yeast and *Hydra* reproduce by budding.
- Confusing pollination (transfer of pollen) with fertilisation (fusion of male and female gametes).
- Saying that fertilisation takes place in the uterus; it takes place in the fallopian tube, and the embryo implants in the uterus.
- Writing that offspring of asexual reproduction show variation; they are identical to the parent (variation arises in sexual reproduction).
- Giving 46 as the number of chromosomes in a human sperm or egg; gametes have 23, the zygote has 46.
- Thinking the sex of the baby is decided by the mother; every egg carries X, so the type of sperm (X or Y) decides.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** zygote	Fertilised egg.
2	**(b)** oviduct (fallopian tube)	Upper part.
3	**(a)** budding	Bud on the body.
4	**(c)** ovules	Become seeds.
5	**(b)** water, air and suitable temperature	Conditions.
6	**(b)** uterus	Wall of uterus.
7	**(b)** spores	Spore-bearing sporangia.
8	**(d)** pollination	Pollination.
9	**(b)** frog and fish	In water.
10	**(a)** placenta	Exchange.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gametes from two parents combine (1); Offspring get a new combination of characters (1).	1+1
12	Growth of a lost body part (1); Planaria or Hydra (1).	1+1
13	Calyx and corolla (1); Androecium and gynoecium (1).	1+1
14	Self: pollen to the stigma of the same flower or plant (1); Cross: pollen to a flower of another plant (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sepals protect the bud; petals attract insects (1); Stamens produce pollen (1); Carpel contains the ovules (1).	1+1+1
16	Testes produce sperms (1); Vas deferens carries sperms (1); Urethra, penis, glands (1).	1+1+1
17	Water, air, temperature (1); Seed swells, radicle emerges (1); Plumule becomes shoot (1).	1+1+1
	OR: Pollen reaches the stigma (1); Pollen tube grows, fertilisation (1); Ovule becomes seed, ovary becomes fruit (1).	1+1+1
18	Ovaries produce ova (1); Oviducts and uterus (1); Vagina (1).	1+1+1
	OR: Zygote: fertilised egg (1); Embryo: early stages of development (1); Foetus: later stages with recognisable features (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pollination (1); Pollen tube (1); Fertilisation (1); Seed and fruit (1); Dispersal by wind, water, animals (1).	1+1+1+1+1
	OR: Self and cross (1); Wind (1); Water and insects (1); Birds and bats (1); Adaptations (1).	1+1+1+1+1
20	Testes (1); Vas deferens and glands (1); Ovaries (1); Oviduct and uterus (1); Hormones (1).	1+1+1+1+1
	OR: Roots, stems and leaves (1); Cutting (1); Layering and grafting (1); Tissue culture (1); Advantages (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that budding is the method of *Amoeba*; *Amoeba* divides by binary fission, whereas yeast and *Hydra* reproduce by budding.
- Confusing pollination (transfer of pollen) with fertilisation (fusion of male and female gametes).
- Saying that fertilisation takes place in the uterus; it takes place in the fallopian tube, and the embryo implants in the uterus.
- Writing that offspring of asexual reproduction show variation; they are identical to the parent (variation arises in sexual reproduction).
- Giving 46 as the number of chromosomes in a human sperm or egg; gametes have 23, the zygote has 46.
- Thinking the sex of the baby is decided by the mother; every egg carries X, so the type of sperm (X or Y) decides.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** regeneration	Growth of lost parts.
2	**(a)** self-pollination	Same flower.
3	**(d)** humans and birds	Inside the body.
4	**(b)** 28 days	Monthly.
5	**(b)** tubers (eyes)	Buds on tubers.
6	**(a)** cross-pollination	Variation.
7	**(c)** 23	46 chromosomes.
8	**(d)** the egg is not fertilised	Lining shed.
9	**(d)** stem cuttings	Vegetative propagation.
10	**(d)** bright coloured and fragrant	Attract insects.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	New plants from vegetative parts (1); Cutting and layering/grafting (1).	1+1
12	Sperm (1); Ovum (1).	1+1
13	Pollination: transfer of pollen to stigma (1); Fertilisation: fusion of male and female gametes (1).	1+1
14	Supplies food and oxygen to the embryo (1); Removes waste (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Number of parents (1); Gametes (1); Variation (1).	1+1+1
16	One ovum released every month (1); Uterus lining thickens (1); If not fertilised, it is shed as menstruation (1).	1+1+1
17	Spore is unicellular, seed is multicellular (1); Seed has a stored food (1); Spore forms by asexual reproduction (1).	1+1+1
	OR: Cutting: rose (1); Layering: jasmine (1); Grafting: mango (1).	1+1+1
18	Amniotic fluid cushions (1); Placenta gives food and oxygen (1); Wastes removed (1).	1+1+1
	OR: Many sperms are lost on the way (1); Ovum is large with stored food (1); One ovum is released each month (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fertilisation in the oviduct (1); Zygote divides (1); Implantation in the uterus (1); Placenta and development (1); Birth after 9 months (1).	1+1+1+1+1
	OR: Frog: external fertilisation (1); Eggs in water, tadpole (1); Hen: internal fertilisation (1); Egg with shell (1); Incubation (1).	1+1+1+1+1
20	Ovulation (1); Thickening of uterine lining (1); Menstruation (1); Pregnancy prevents it (1); 28-day cycle (1).	1+1+1+1+1
	OR: Causes: high birth rate (1); Effects on resources (1); Contraceptives (1); Awareness (1); Government schemes (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that budding is the method of *Amoeba*; *Amoeba* divides by binary fission, whereas yeast and *Hydra* reproduce by budding.
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